PROJECT <A>

2026 Mini Project Summary

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One team, four specialties

Backend · AI · Frontend · The infrastructure is connected as a single OCR service.

01 BACKEND

Kim Youngmin

- API development based on Spring Boot and integration with React–FastAPI OCR
- Implementing JWT-based user authentication
- Structure of the Document Upload and OCR Result Processing Flow
- MySQL integration and implementation of data storage structure
- Mockito unit tests and preparation of test plans and results
- Creating a project plan screen definition

02 PM · INFRA

Choi Geonwoo

- GitHub organization, repository, and project resource management
- Operation of Docker image and container health check system
- .env-based organizational unit environment variables and deployment settings
- Distribution of OA materials and operation of live services
- Web UI configuration

From backend → AI → frontend → infrastructure operations, we implemented a single end-to-end OCR experience.

One team, four specialties

Backend · AI · Frontend · The infrastructure is connected as a single OCR service.

03 AI MODEL

Choi Minki

- FastAPI OCR Server Development and AI Model Integration
- OCR API unit testing and pytest test code writing

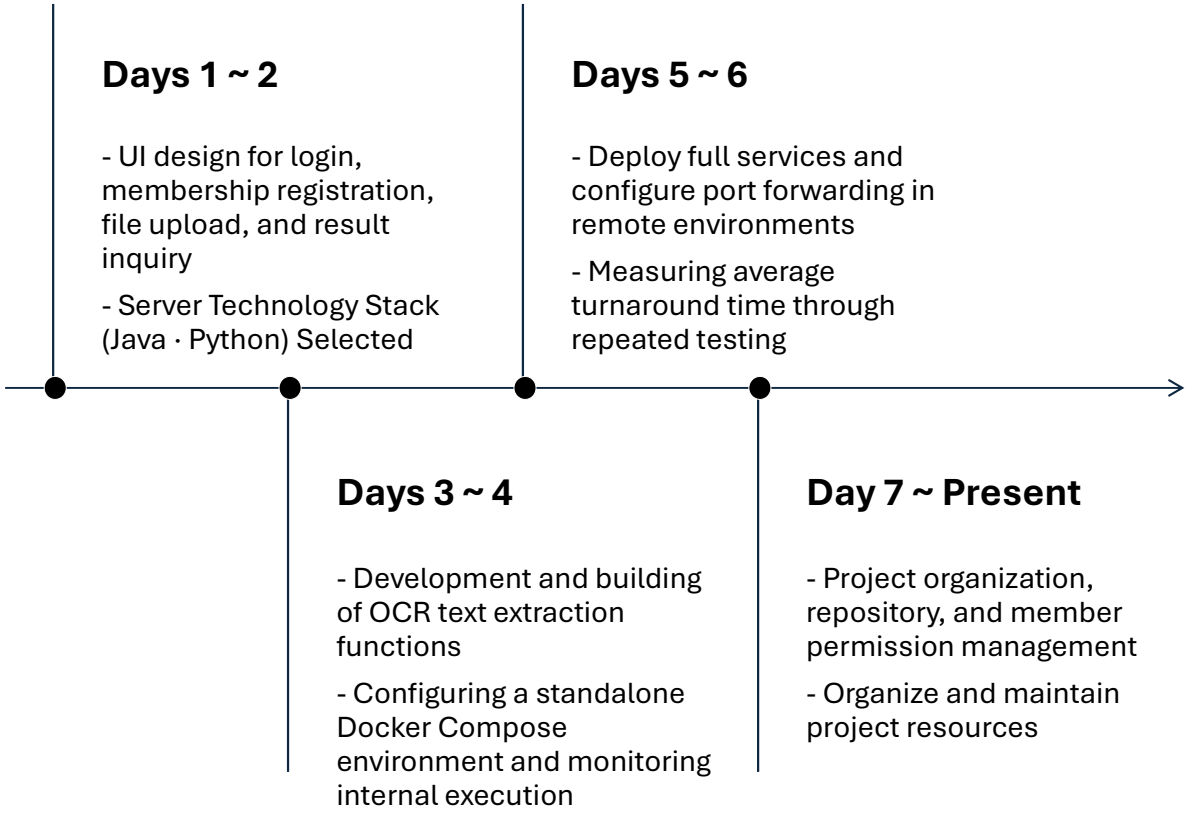
04 FRONTEND

Yoon Seungjin

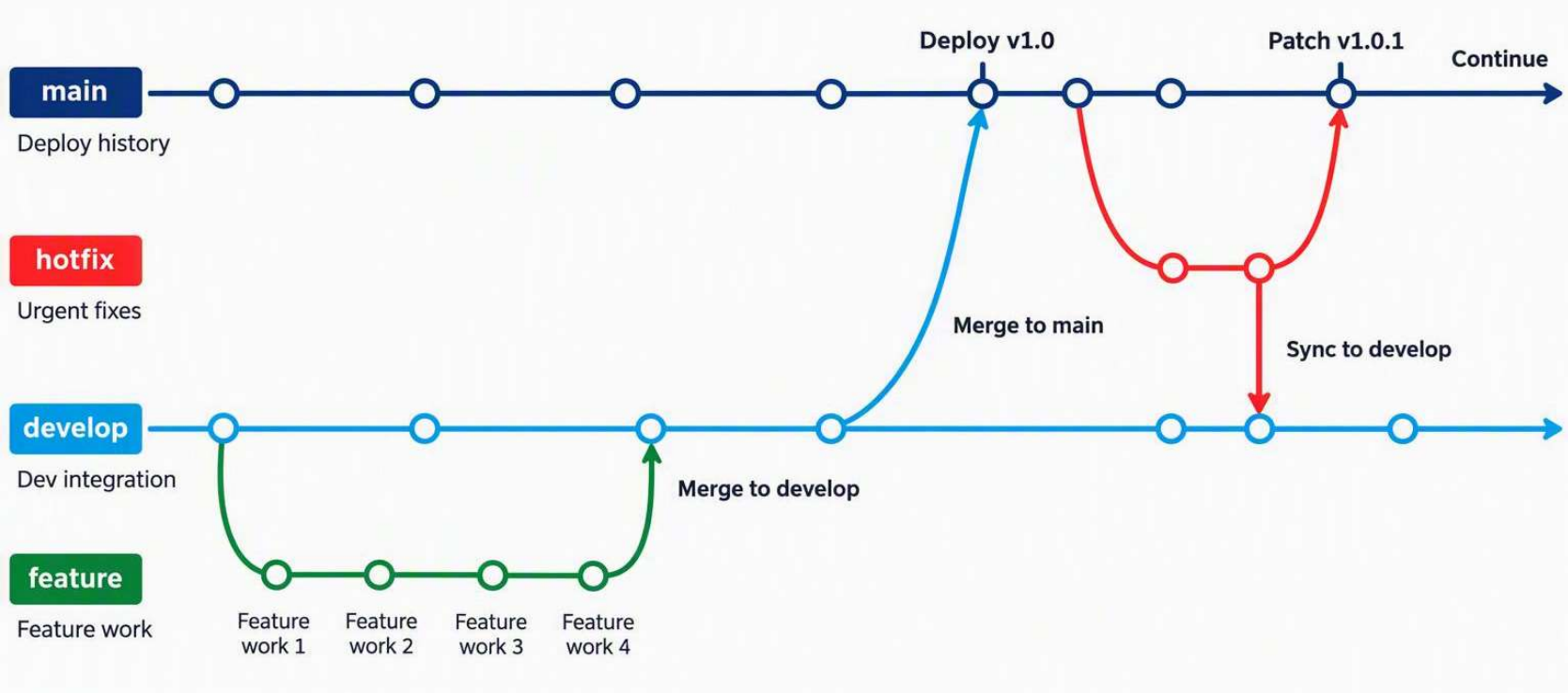
- Implementation of React/Vite-based OCR document management screens
- API integration and user event handling logic development

From backend → AI → frontend → infrastructure operations, we implemented a single end-to-end OCR experience.

Development Schedule

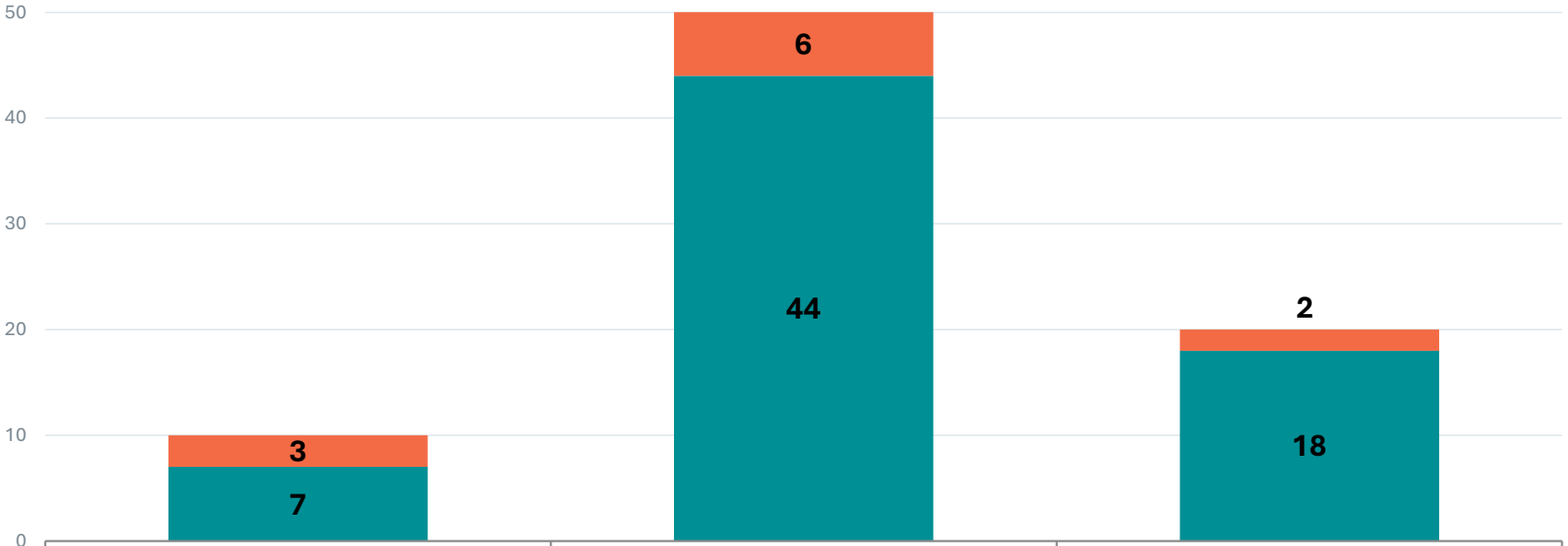


Development Branching Strategy



69 out of 80 PRs approved

Approvals by **repository** · **Cancellation status** | Server 50 records · Web 20 records · Infra 10 records



	Infra	Server	Web
cancellation	3	6	2
Approval	7	44	18

Automate everything from PR creation to collaborative sharing

- **GitHub Actions supports quickly sharing development status** by applying language-specific PR templates tailored to user nationalities

AUTOMATION FLOW

01 Verify User Information

Searching for nationality information via GitHub API

02 Applying Templates by Language

Automatic selection of PR templates tailored to nationality

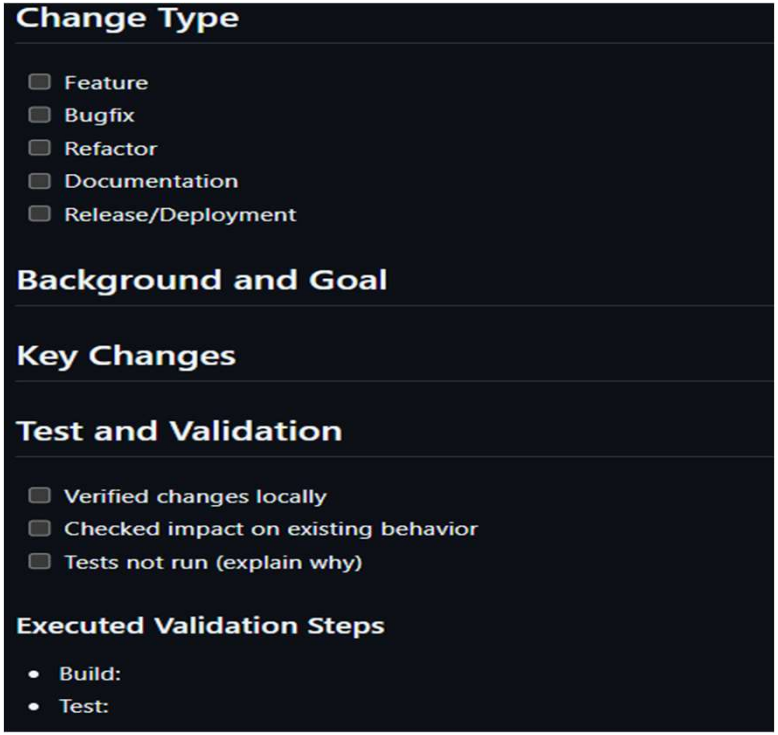
03 Change Summary Expansion

Major changes based on Commit SHA-256 are expected to be reflected.

Summary of changes → applying the nationality verification → template

Automating PR writing rules to accelerate collaboration sharing

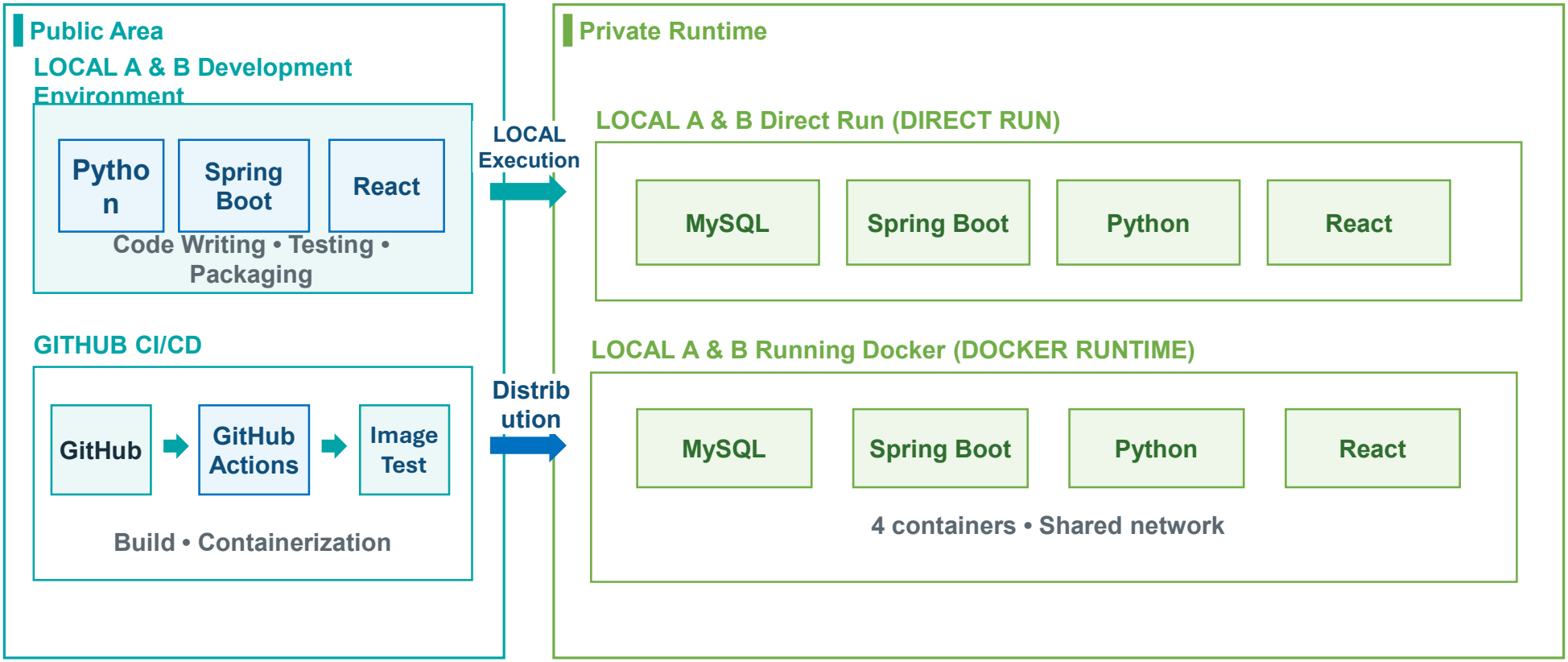
PR TEMPLATE | Automation Application Screen



Architecture & Infrastructure

Presenter: Choi Geon-woo

Architecture



Development → Build → Containerization → Deployment → Execution

Organizational unit deployment environment variable value management

01 • Define repository variables

```
# Environment variables for the server services
WEB_FRONTEND_PORT=YOUR_WEB_FRONTEND_PORT
GO_SERVER_PORT=YOUR_GO_SERVER_PORT

# PostgreSQL environment variables for the project
POSTGRES_DB=YOUR_DATABASE_NAME
POSTGRES_USER=YOUR_DATABASE_USER
POSTGRES_PASSWORD=YOUR_DATABASE_PASSWORD
POSTGRES_HOST=YOUR_DATABASE_HOST
POSTGRES_PORT=YOUR_DATABASE_PORT

# MySQL environment variables for the project
MYSQL_DATABASE=YOUR_DATABASE_NAME
MYSQL_USER=YOUR_DATABASE_USER
MYSQL_PASSWORD=YOUR_DATABASE_PASSWORD
MYSQL_ROOT_PASSWORD=YOUR_DATABASE_PASSWORD
MYSQL_HOST=YOUR_DATABASE_HOST
MYSQL_PORT=YOUR_DATABASE_PORT
```

```
# Python server environment variables
PYTHON_SERVER_PORT=YOUR_PYTHON_SERVER_PORT
APP_NAME=YOUR_APP_NAME
APP_VERSION=YOUR_APP_VERSION
APP_ENV=YOUR_APP_ENV
DEBUG=YOUR_DEBUG
AI_PROVIDER=YOUR_AI_PROVIDER

OPENAI_API_KEY=YOUR_OPENAI_API_KEY
OPENAI_MODEL=YOUR_OPENAI_MODEL

PDF_BACKEND=YOUR_PDF_BACKEND
DEFAULT_OCR_ENGINE=YOUR_DEFAULT_OCR_ENGINE

AI_TIMEOUT_SECONDS=YOUR_AI_TIMEOUT_SECONDS
AI_MAX_OUTPUT_TOKENS=YOUR_AI_MAX_OUTPUT_TOKENS

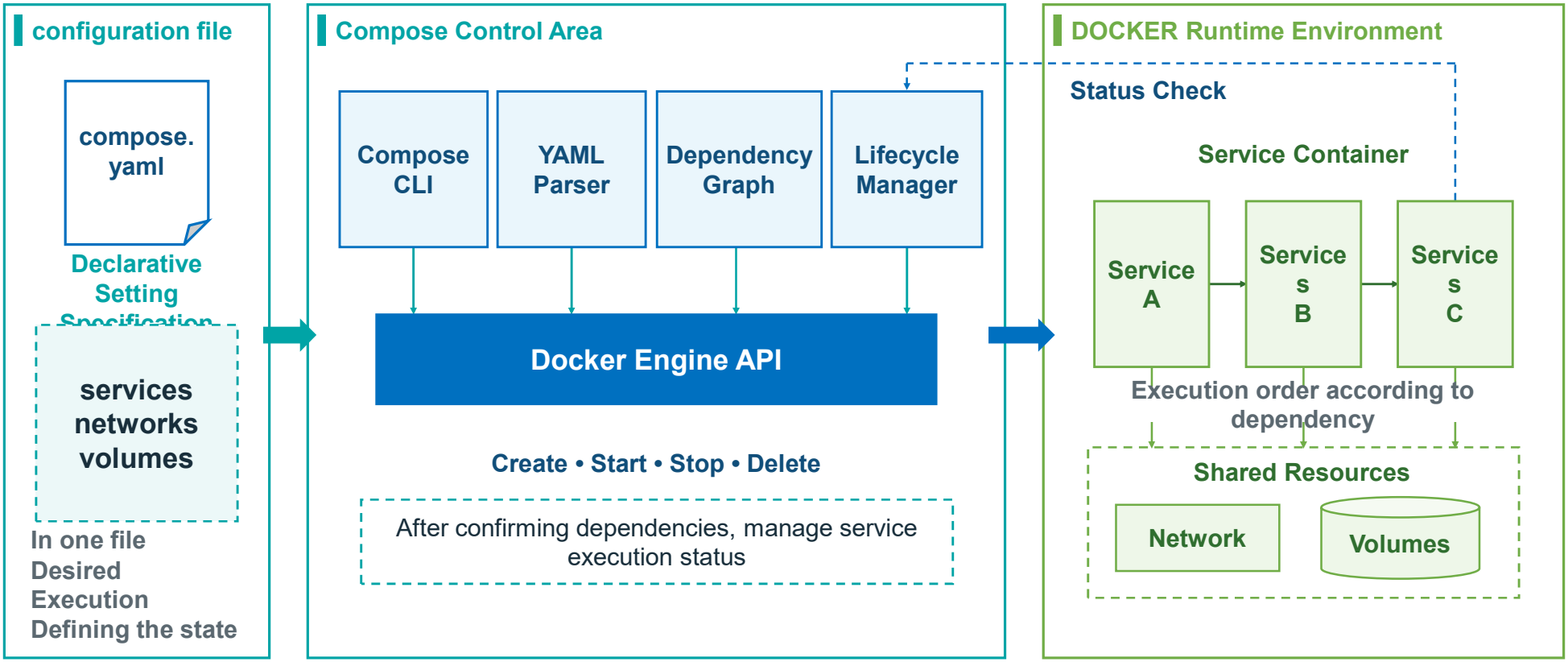
NETWORK_ENV=YOUR_NETWORK_ENV
ENGINE=YOUR_ENGINE
```

02 • Register your organization's secret



Variable definition → organizational unit registration → consistent use in deployment environments

Architecture



Declaring settings → interpreting settings → checking dependencies → executing control → running services

Running Docker Compose-Based Services

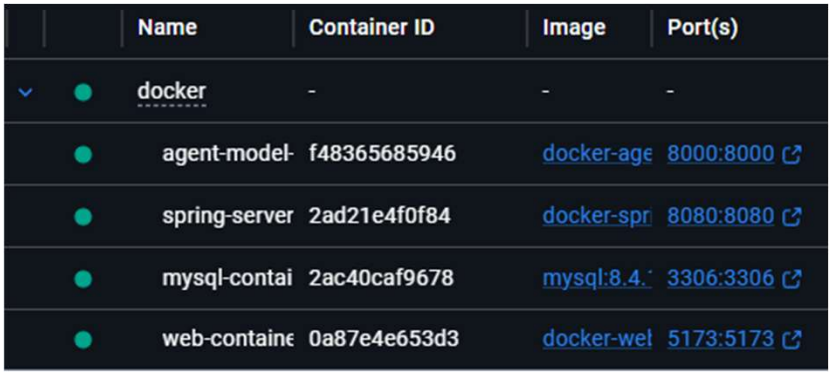
01 • Defining Execution Configuration

```
# MySQL Database Service
▷Run Service
ocr-mysql:
  container_name: mysql-container
  image: mysql:8.4.11-oraclelinux9
  shm_size: '256mb'
  restart: always
  volumes:
    - ../db/schema.sql:/docker-entrypoint-initdb.d/schema.sql:ro
    - mysql-data:/var/lib/mysql
  environment:
    MYSQL_HOST: ${MYSQL_HOST}
    MYSQL_DATABASE: ${MYSQL_DATABASE}
    MYSQL_USER: ${MYSQL_USER}
    MYSQL_PASSWORD: ${MYSQL_PASSWORD}
    MYSQL_ROOT_PASSWORD: ${MYSQL_ROOT_PASSWORD}
  ports:
    - "${MYSQL_PORT}:3306"
  healthcheck:
    test: ["CMD-SHELL", "mysqladmin ping -h localhost -uroot -p$$MYSQL_ROOT_PASSWORD --silent"]
    interval: 10s
    timeout: 5s
    retries: 20

volumes:
  mysql-data:
    driver: local
```

→

02 • Container Status Check



	Name	Container ID	Image	Port(s)
⌄	docker	-	-	-
	agent-model-	f48365685946	docker-age	8000:8000
	spring-server	2ad21e4f0f84	docker-spr	8080:8080
	mysql-contai	2ac40caf9678	mysql:8.4.	3306:3306
	web-containe	0a87e4e653d3	docker-wel	5173:5173

Define Compose → Run service containers

React Web

Presenter: Yoon Seung-jin

Components handle common functions, Pages handle user flow

Reuse, authentication control, and screen-by-screen input, API processing, separated structure

SHARED LAYER

Components/

UI and authentication logic shared across multiple screens

`protected-route.jsx`

Token verification · Protected screen access control

Processing the common authentication flow before entering the screen

`feature-header.jsx`

Brand header · Logout event

Reusing the same UI and actions across multiple screens

ROUTE LAYER

Pages/

Real screens and user events at the URL level

`signin.jsx · register.jsx`

Login · Sign up · API call

`feature.jsx · upload.jsx`

Document List · PDF Upload · Engine Selection

`document-detail.jsx`

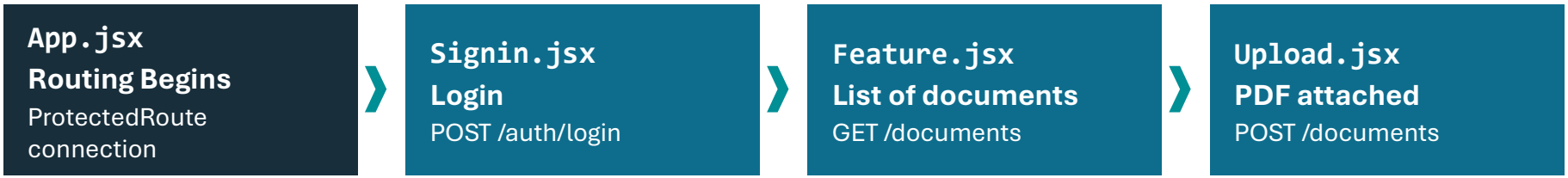
Polling · Displaying Results · Reprocessing · Deletion · Export

Combining cross-functional functions into Components, while Pages focuses on screen-by-screen tasks.

App routing connects the flow of pages, and common components support the protected screen

From logging in → document list → PDF upload, connecting screen transitions and API calls in one flow

ROUTE FLOW | Pages/



Common application to protective screens

SHARED LAYER | Components/

`protected-route.jsx`

Verification before entering the protection screen

`feature-header.jsx`

Common header · Logout processing

Core Logic | App Routing → Pages Processing → Components Authentication · UI Reuse

Spring Boot Server

Presenter: Kim Young-min

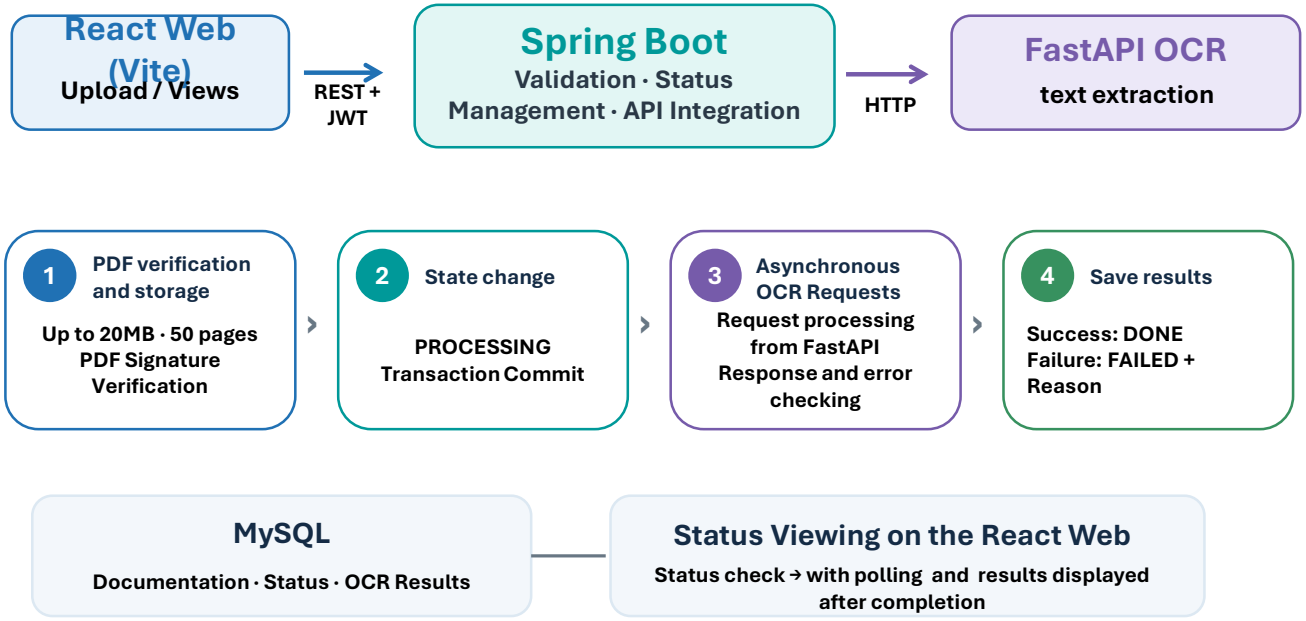
Spring Boot Server Roles and Processing Flow

01 JWT Certification

02 PDF verification and storage

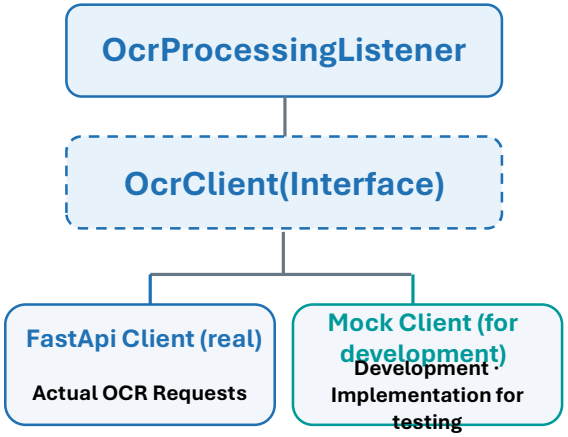
03 Managing Processing Status

04 Save results



Design Points and Future Expansion Directions

01 Replaceable OCR Client



• Switching implementation to settings (app.ocr.client)

02 Unit Testing and Exception Handling

161 items

Unit Test Case

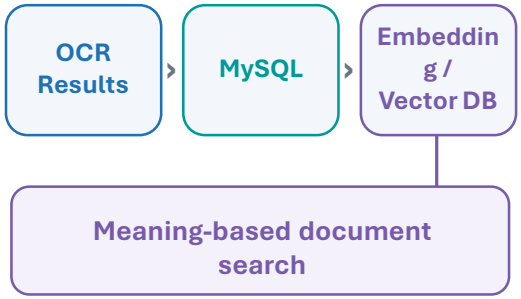
DB · Network-free (Mookito-based verification)

- File Boundary Value
- Certification
- OCR Flow
- error response

• Timeout and error mapping

03 Future Expansion: Vector DB

Maintaining the existing structure while adding steps



• Future expansion with RAG capabilities

AI Model

Presenter: Choi Min-ki

Choose OCR models based on processing speed and precision

Common Criteria Open Source License MIT · Apache-2.0 | Supported languages: English · Korean

SPEED / VOLUME

Tesseract

LSTM-based from version 4.0

- Relatively lightweight and fast on CPU
- Advantageous for quickly handling large volumes of documents

PRECISION / COMPLEX INPUT

PaddleOCR

Full-process deep learning-based

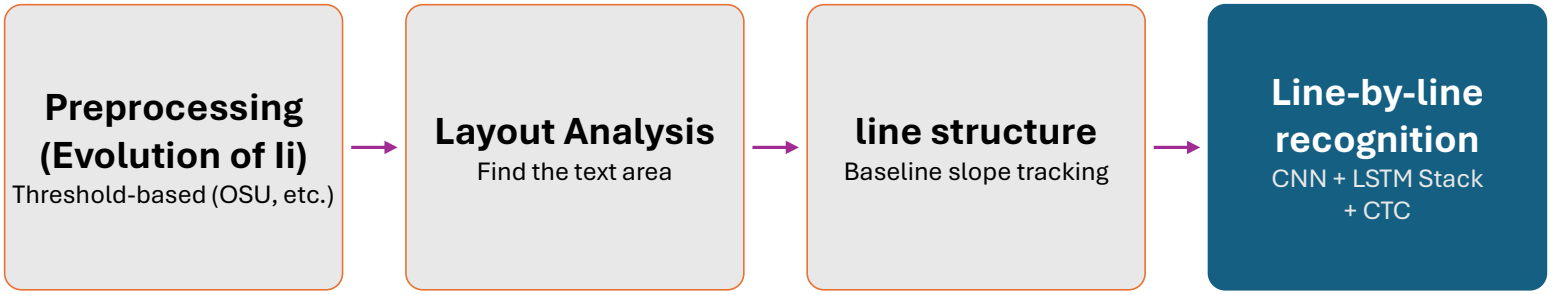
- CPU environment threads leveraged, GPU acceleration possible
- Glass for precise recognition of complex images

AI Model Logic

Processing flow from input image to text output

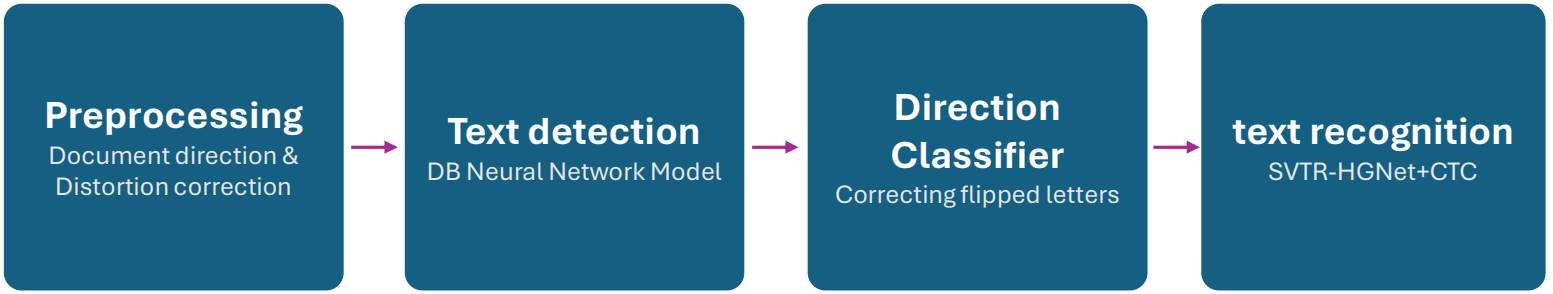
Tesseract

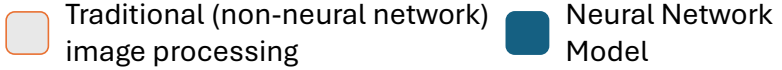
Non-neural network-based frontend
+ LSTM recognition



PaddleOCR

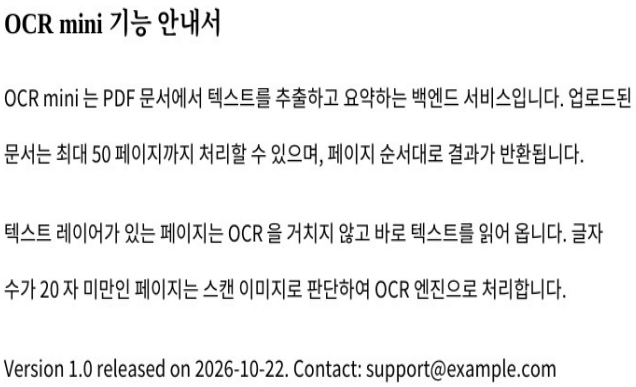
Step-by-step deep learning
Model Connection





AI Model Benchmarking

Normal PDF



OCR mini 기능 안내서

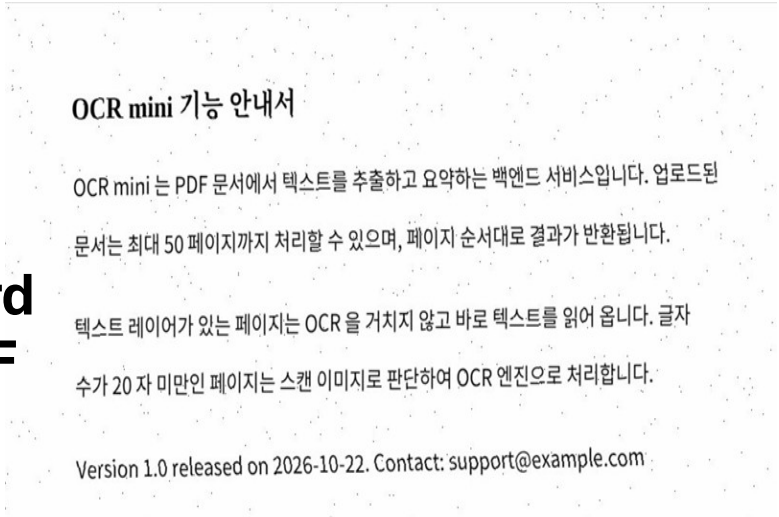
OCR mini 는 PDF 문서에서 텍스트를 추출하고 요약하는 백엔드 서비스입니다. 업로드된 문서는 최대 50 페이지까지 처리할 수 있으며, 페이지 순서대로 결과가 반환됩니다.

텍스트 레이어가 있는 페이지는 OCR 을 거치지 않고 바로 텍스트를 읽어 옵니다. 글자 수가 20 자 미만인 페이지는 스캔 이미지로 판단하여 OCR 엔진으로 처리합니다.

Version 1.0 released on 2026-10-22. Contact: support@example.com

	CER(%)	Accuracy (%)	Processing Time (Ms)
Tesseract	34	66	7491
Paddle	4	96	32455

Hard PDF



OCR mini 기능 안내서

OCR mini 는 PDF 문서에서 텍스트를 추출하고 요약하는 백엔드 서비스입니다. 업로드된 문서는 최대 50 페이지까지 처리할 수 있으며, 페이지 순서대로 결과가 반환됩니다.

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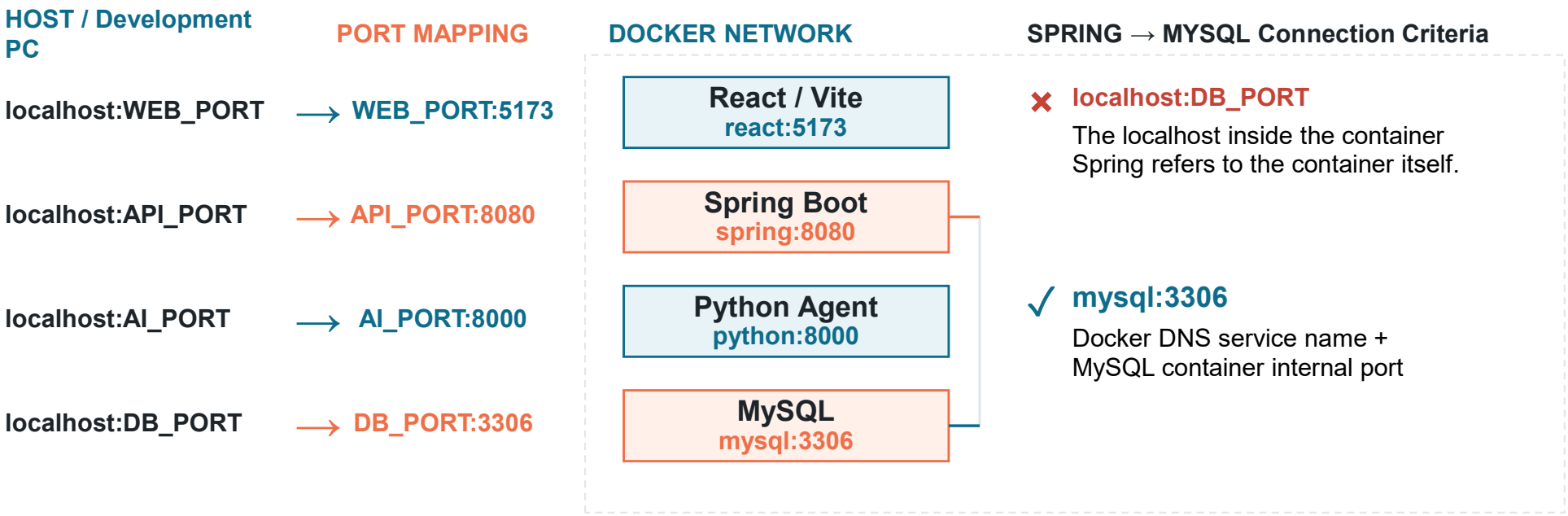
Version 1.0 released on 2026-10-22. Contact: support@example.com

	CER(%)	Accuracy (%)	Processing Time (Ms)
Tesseract	38	62	7016
Paddle	10	90	16881

Problem Solving

Container Port Mapping Troubleshooting

HOST_PORT is used for external public disclosure, and inter-service communication uses Docker servicename + CONTAINER_PORT.



Causes Host public port and container internal port Using the same reference standard

Solution Spring: DB_HOST=mysql · DB_PORT=3306 External access only localhost:DB_PORT

Future Development Plans

Summary

Project Summary

We connected the entire development, deployment, and OCR workflow. Now is the time to improve operational quality and scalability.

Project Summary

01 Collaboration and Distribution

We integrated the branching strategy with GitHub Actions to build the flow from code changes to build, testing, and containerization.

02 Production Preparation Architecture

Python, Spring Boot, React, and MySQL all support direct execution and Docker runtime, and all have a compose-based lifecycle.

03 End-to-End OCR Applications

Through user requests and API validation, we implemented core workflows through agent preprocessing, tesseract inference, and MySQL storage.

Expected Roadmap

Step 1 — **Operational Stability**

Integrated test expansion · Improved error handling · Performance baseline definition

Step 2 — **Automation and Observability**

Strengthening CI/CD quality gates · Improved visibility based on logs and metrics

Step 3 — **Scale and Optimization**

Scalable infrastructure assessment · OCR/AI model improvements · Service expansion

Related links



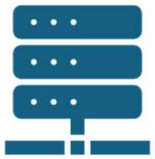
Team Member Profiles

geonwoo8873 (GEONWOO CHOI)

GreyColor10 (Mingi Choi)

Kymin9118

tmdwls1226



Project Repository

geonwoo8873/global-academy-project [Private]

Org-global-academy-project



References

geonwoo8873/custom-repo-template

Hugging Face – The AI community building the future.

Vite | Next Generation Frontend Tooling

W3Schools Online Web Tutorials

wingetCollections

THANK YOU

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